

Challenge 01 - Wireshark Filtering



Exam-Simulation-Challenge01.pcap

• Task_1- Attack Type

Which kind of attack was detected in the network traffic dump?

The analysis of the provided PCAP network traffic indicates a DNS tunneling attack, most likely used for covert data exfiltration or command-and-control (C2) communication.

• Task_2

Based on what characteristics the attack can be detected?

The attack was detected based on several abnormal DNS traffic characteristics observed in the capture:

- **Unusually high volume of DNS queries** originating from a single internal host
- **Excessively long DNS query names**, significantly longer than those seen in normal DNS usage
- **High-entropy, random-looking subdomain labels**, indicating encoded data rather than human-readable domain names
- **Repeated DNS queries to the same external domain**, suggesting an automated process
- **Consistent timing and structure** of the DNS requests, indicative of malware behavior rather than legitimate user activity

• Task_3

What is the exact starting point (packet number) of the attack?

The exact starting point of the attack : Frame 27

• Task_4

What operating system (not version) and source IP the attack was originating from?

- Source IP: 192.168.5.10
- Based on a google search for TTL (128): Windows

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• Task_5

What was the motivation for the attack and which goal most likely was to be achieved?

Given that Batnet Laboratories Inc. is involved in military-related projects, the most likely motivation behind the attack was espionage or data theft.

Most probable goals:

- Covert data exfiltration using DNS as a stealth communication channel
- Command-and-control communication between malware inside the network and an external attacker-controlled server